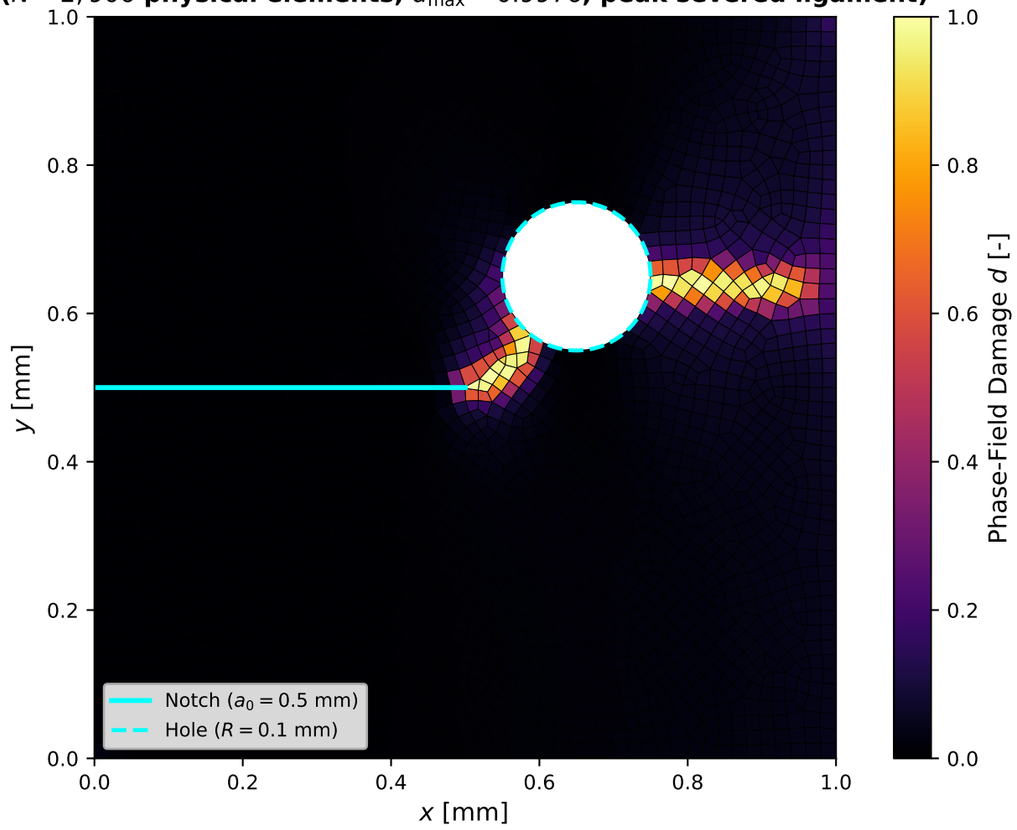
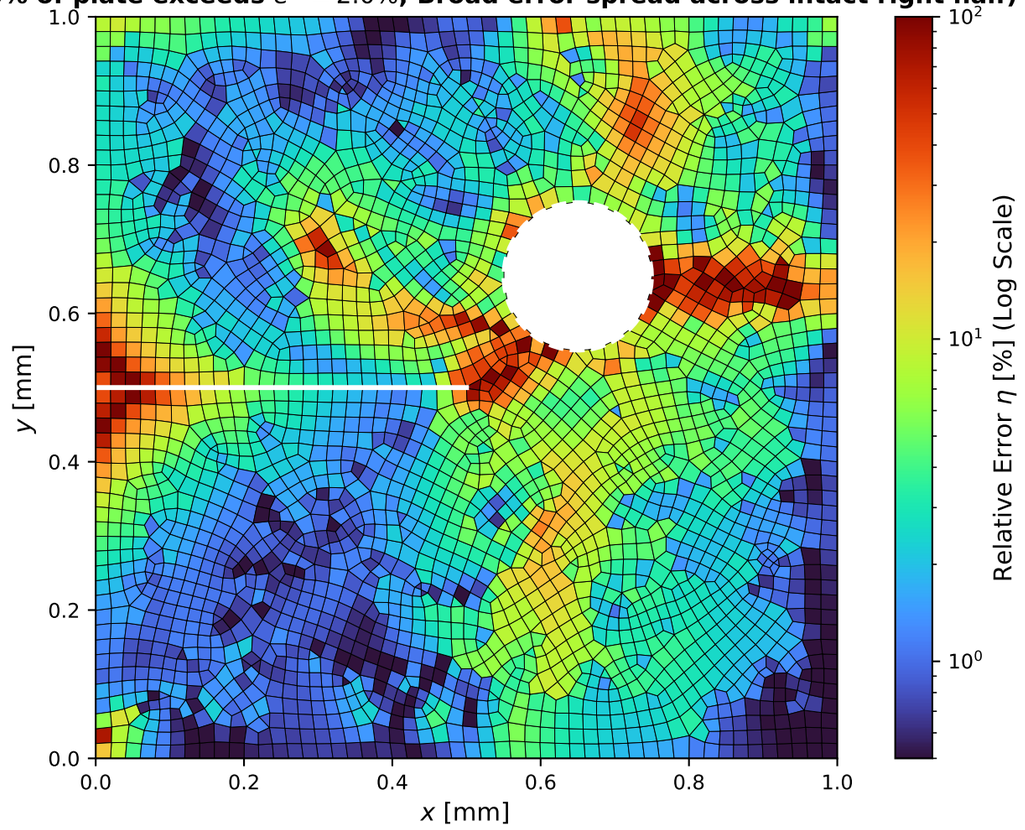
(a) Authentic Donor Phase-Field Damage $d_{\max}$ (Job 1 Terminal State)
($N = 2,906$ physical elements, $d_{\max} = 0.9976$, peak severed ligament)

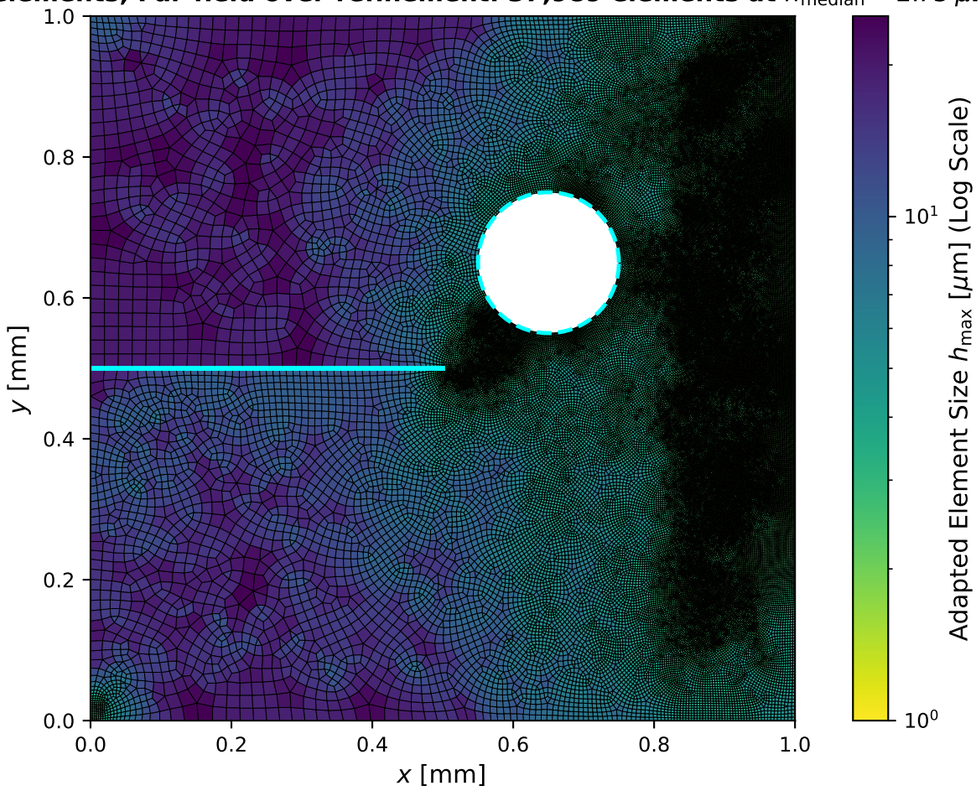


(b) Recovery Relative Error Indicator $\eta = (\text{MISESERI}/\text{MISESAVG}) \times 100\%$
(59.25% of plate exceeds $e^* = 2.0\%$; Broad error spread across intact right half)



85.1%

(c) Resulting Adapted Mesh Resolution $h_{\max}$ [μm]
(77,211 elements; Far-field over-refinement: 37,989 elements at $h_{\text{median}} = 2.73$ μm)



(d) Quantitative Causal Mismatch: Far-Field Cost vs Core Under-Allocation
(Undamaged far field consumes 49.2% of mesh elements due to non-zero relative error)

